

LECTURE 12

CYBERSAFETY



LECTURE CONTENT



- Information system security
- Identification, Authentication, Authorization
- Strong password requirements
- Information security in computer networks
- Brannauer (firewall)
- Cryptographic protection of information
- Encryption and decryption
- Encryption algorithms
- Electronic digital signature
- Computer viruses and malware
- Antivirus programs
- Phishing
- Information security standards (ISO)

INFORMATION SYSTEM SECURITY

The security of an information system is a property that consists in the ability of a system to ensure its normal functioning, that is, to ensure the integrity and secrecy of information.





Sources of threats to the security of information systems:

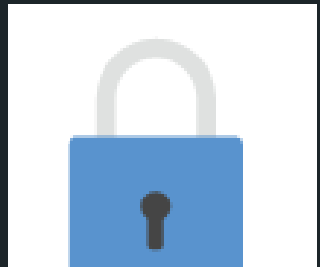
- anthropogenic sources caused by accidental or deliberate actions of subjects;
- technogenic sources leading to failures of hardware and software or software errors;
- spontaneous sources caused by natural disasters or force majeure.

POSSIBLE DIRECTIONS OF INFORMATION LEAKAGE AND WAYS OF UNAUTHORIZED ACCESS:

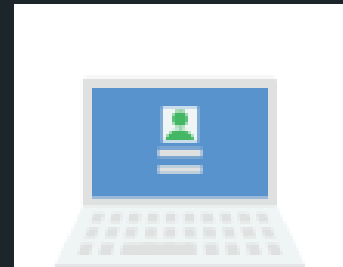


- interception of information;
- modification of information (the original message or document is changed or replaced by another and sent to the addressee);
- substitution of authorship of information (someone can send a letter or document on your behalf);
- exploiting the shortcomings of operating systems and application software;
- copying of data carriers and files with overcoming protection measures;
- disguise as a registered user and assigning his powers;
- the introduction of computer viruses and so on.

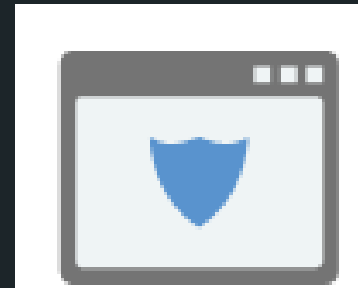
THE MEANS OF PROTECTING INFORMATION FROM THE ACTIONS OF SUBJECTS



means of
protecting
information from
unauthorized
access



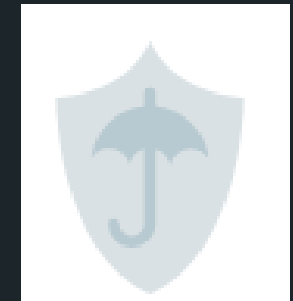
computer networks
security



cryptography
techniques in
information
security

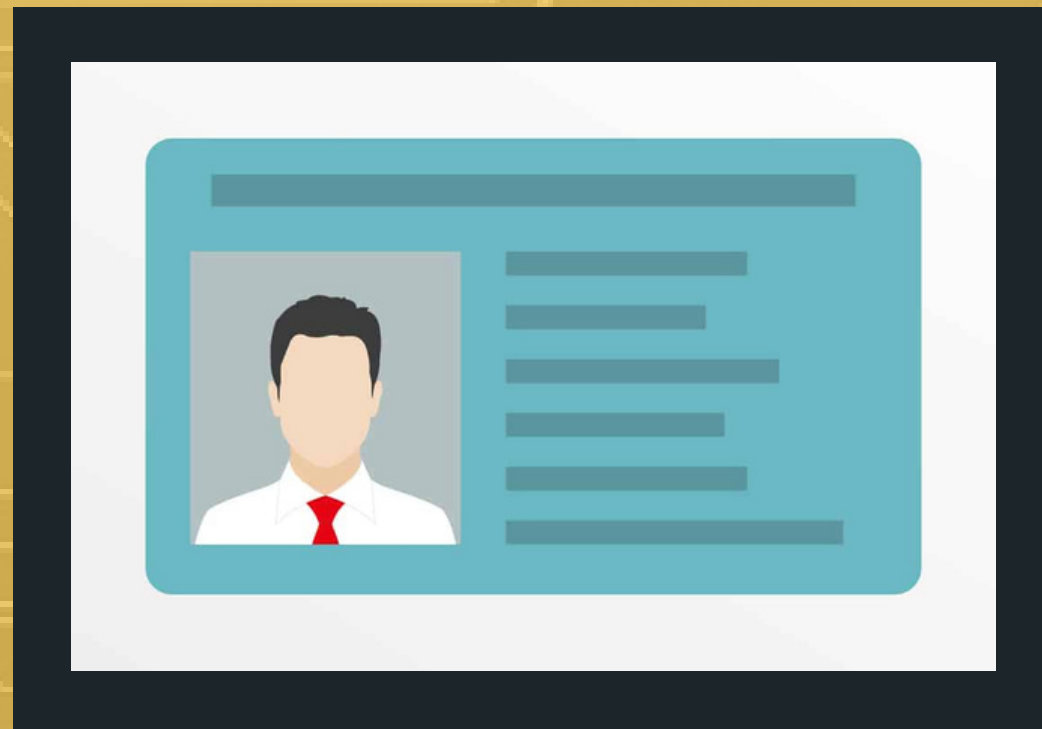


electronic digital
signature



protection of
information from
computer viruses

Ways to protect information from unauthorized access:



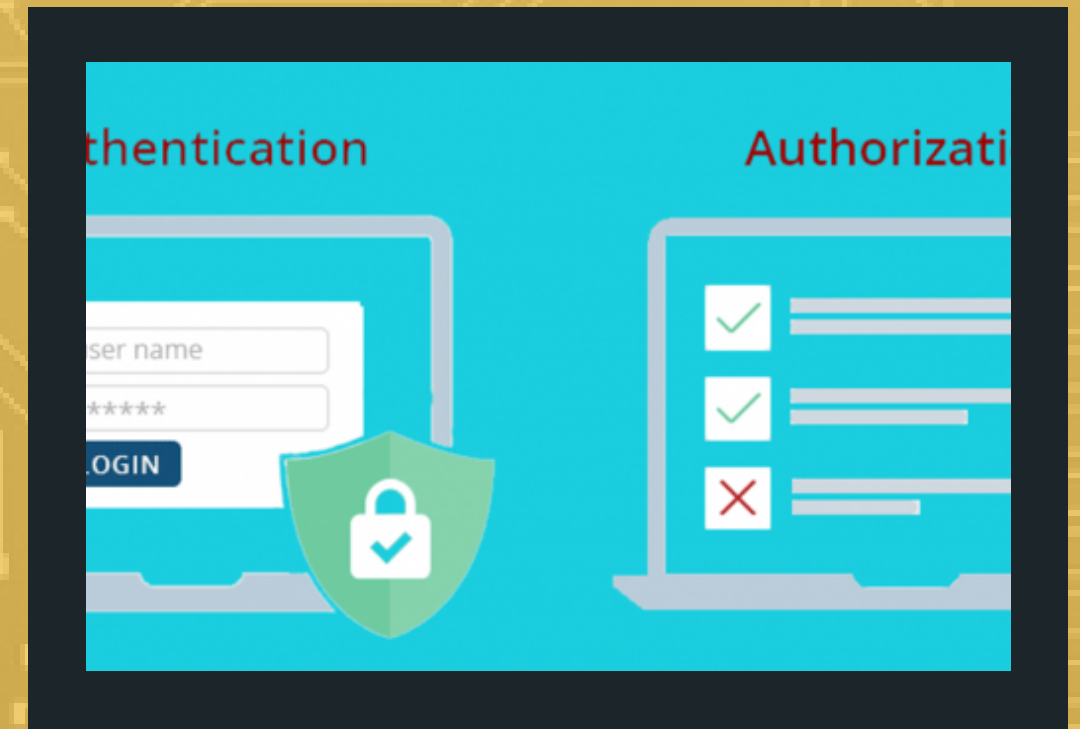
IDENTIFICATION

is the process of recognizing a user by his identifier



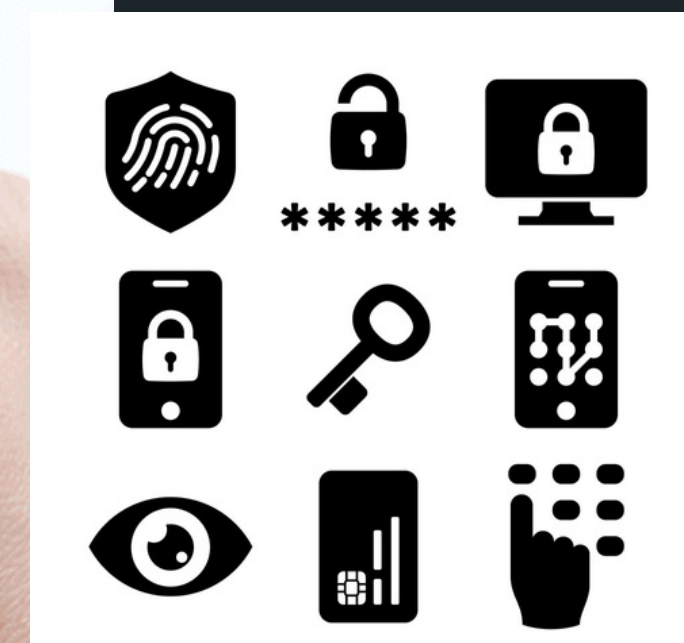
AUTHENTICATION

is an authentication procedure, proof that the user is who he claims to be



AUTHORIZATION

checking the authority or checking the user's right to access specific resources and perform certain operations on them



A STRONG PASSWORD REQUIREMENTS:

- contain at least 8-9 characters;
- a mixture of both uppercase and lowercase letters;
- a mixture of letters and numbers;
- inclusion of at least one special character, e.g., ! @ # ?]

Note: do not use < or > in your password, as both can cause problems in Web browsers.



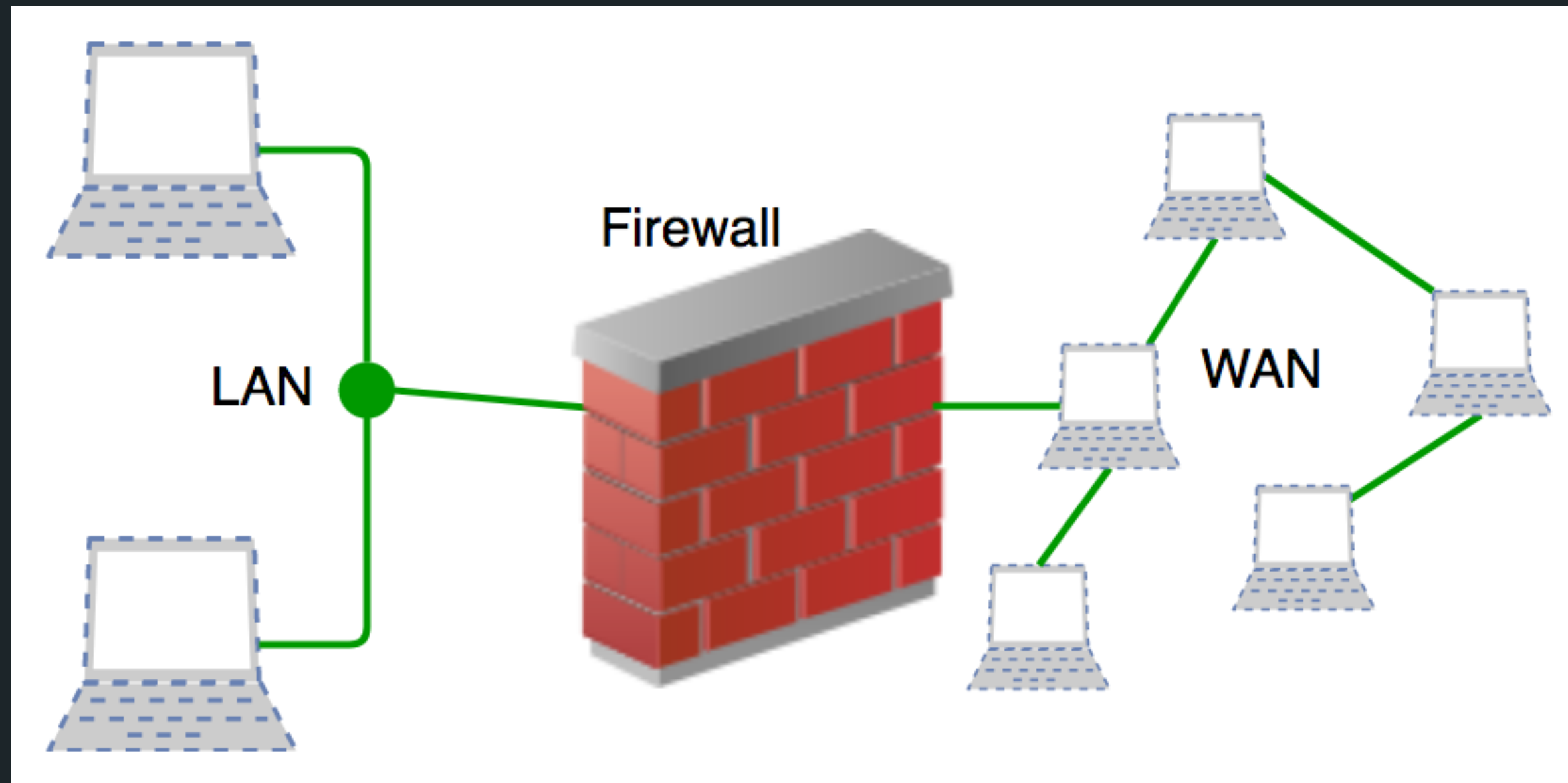
EXAMPLES OF WEAK PASSWORDS:

- Any word that can be found in a dictionary, in any language (e.g., airplane)
- A dictionary word with some letters simply replaced by numbers (e.g., a1rplan3)
- A repeated character or a series of characters (e.g., AAAAA or 12345)
- A keyboard series of characters (e.g., qwerty)
- Personal information (e.g., birthdays, friends' names, addresses, etc.).



FIREWALL

A firewall is a protective screen between the global Internet (WAN) and the local area network (LAN) of an organization. It performs the function of checking and filtering data from the Internet.



Firewalls will be useful for the following organizations:

- Schools and other educational institutions
- Cafes, restaurants and other establishments that distribute Wi-Fi to their visitors.
- Hotels and motels
- Organizations that follow up so that employees are not distracted by non-profile matters. For example, Traffic Inspector Next Generation can block applications like Skype and BitTorrent, resources that do not have any relationship to work, as well as advertising on sites and other harmful content. It also keeps all the statistics - what employee visited what sites and how much time spent on those or other resources.

Cryptology

```
graph TD; A[Cryptology] --> B[Cryptography]; A --> C[Cryptanalysis];
```

A hierarchical diagram with a dark blue background and yellow bars. At the top center is a yellow rectangular box containing the word "Cryptology". Two yellow arrows originate from the bottom corners of this box and point downwards and outwards. The left arrow points to a yellow rectangular box on the bottom left containing the word "Cryptography". The right arrow points to a yellow rectangular box on the bottom right containing the word "Cryptanalysis".

Cryptography

Cryptanalysis

ENCRYPTION GOALS



CONFIDENTIALITY

With cryptographic protection, information can be made inaccessible to outsiders.

IMMUTABILITY

The encrypted information cannot be tampered with or corrupted during transmission or storage.

CONFIRMATION OF THE SOURCE

Encrypted information has properties that confirm that it was sent by a specific person and not by someone else.

Encryption algorithms

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graph TD; A[Encryption algorithms] --> B[Symmetric]; A --> C[Asymmetric];
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Symmetric

same key is used for encryption
and decryption

Asymmetric

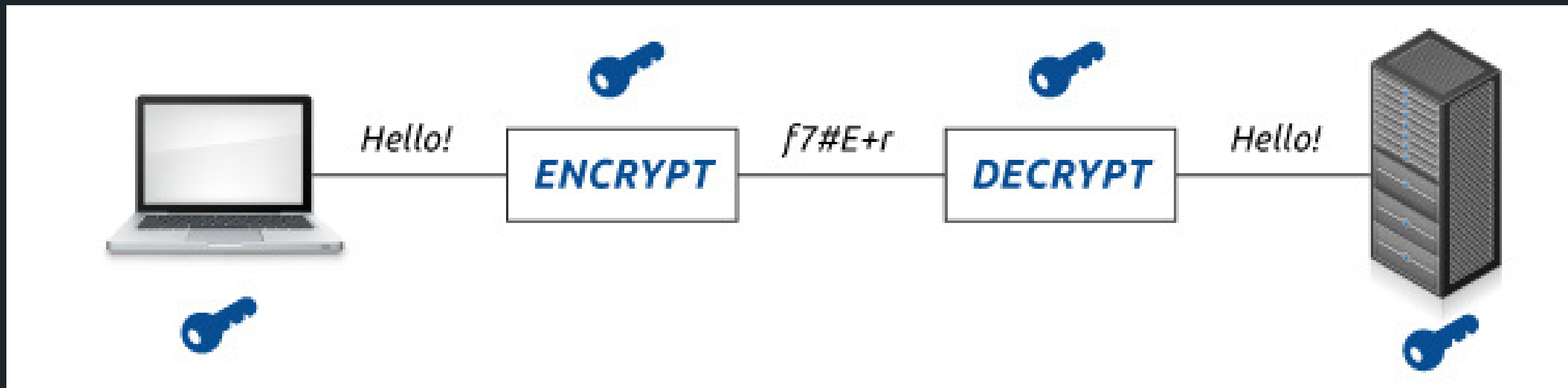
Two keys (public and private)
are used for encryption and
decryption.

THE MOST COMMON ENCRYPTION ALGORITHMS:



Symmetrical: AES, Grasshopper, Camellia, Twofish, Blowfish;

Asymmetric: Elgamal, RSA;



AN ELECTRONIC DIGITAL SIGNATURE

An electronic digital signature is a software and cryptographic tool that provides:

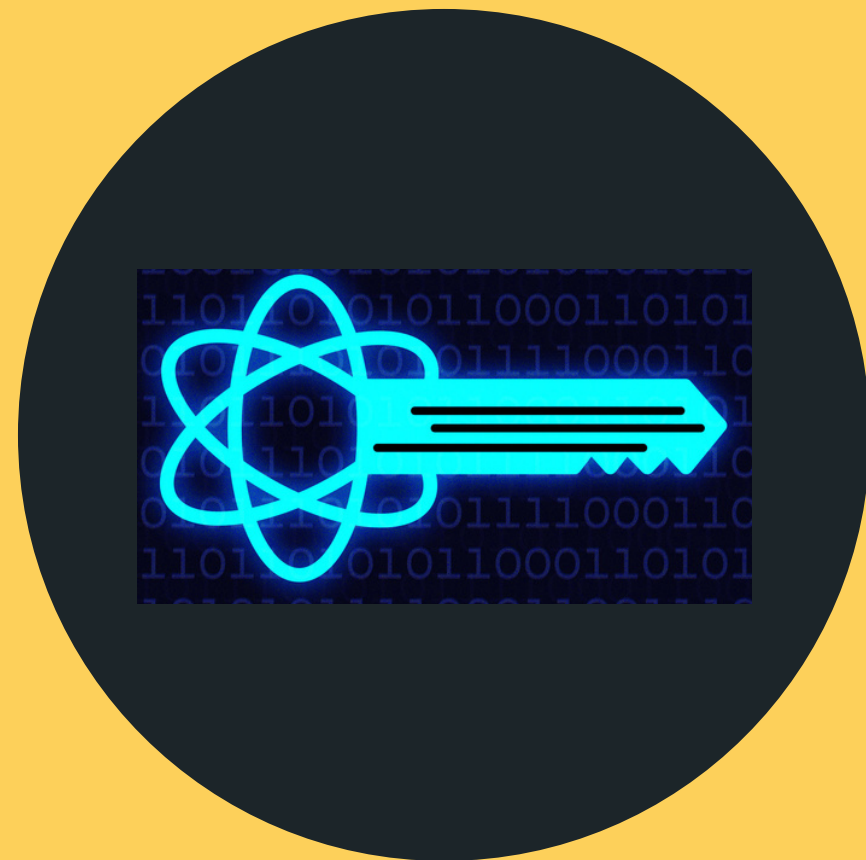
- checking the integrity of documents;
- confidentiality of documents;
- identification of the person who sent the document.



Digital
Signatures



Electronic
Signatures



BENEFITS OF USING DIGITAL SIGNATURE:

- significantly reduce the time spent on registration of the transaction and exchange of documentation;
- improve and reduce the cost of preparation, delivery, accounting and storage of documents;
- guarantee the accuracy of the documentation;
- minimize the risk of financial losses by increasing the confidentiality of information exchange;
- build a corporate document exchange system.

COMPUTER VIRUSES

There are many different ways of classifying computer viruses. One of the ways to classify computer viruses is to separate them according to the following main characteristics:

- habitat;
- features of the algorithm;
- methods of infection;
- degree of exposure (harmless, dangerous, very dangerous).



Depending on the habitat, the main types of computer viruses are:

SOFTWARE VIRUSES

Software viruses are malicious code that is embedded inside executable files (programs). Viral code can reproduce itself in the body of other programs - a process called reproduction.

BOOT VIRUSES

they do not infect program files, but the boot sector of magnetic media (floppy and hard disks)

MACRO VIRUSES

infect documents that are created in application programs that have tools for executing macros.

NETWORK VIRUSES

Network viruses are sent from computer to computer using computer networks, e-mail, and other channels for their distribution.

ACCORDING TO THE ALGORITHMS OF WORK, COMPUTER VIRUSES ARE DISTINGUISHED:

- worms sent from computer to computer via computer networks, e-mail and other channels.
- invisible viruses (stealth viruses);
- trojans;
- programs - mutants;
- logic bombs, etc.



THE MAIN SIGNS OF A VIRUS ON THE PC



slow computer work

freezes and malfunctions of the computer

resizing files

reducing the size of free RAM

a significant increase in the number of files on disk

disappearance of files and directories or distortion of their contents

Anti-virus software



Avast



Bitdefender



AVG



Avira



Kaspersky



Panda

PHISHING

Phishing is a type of Internet fraud, the purpose of which is to obtain user identification data. This includes stealing passwords, credit card numbers, bank accounts and other confidential information. Phishing is a type of fake mail sent to banks, providers, payment systems and other organizations.

